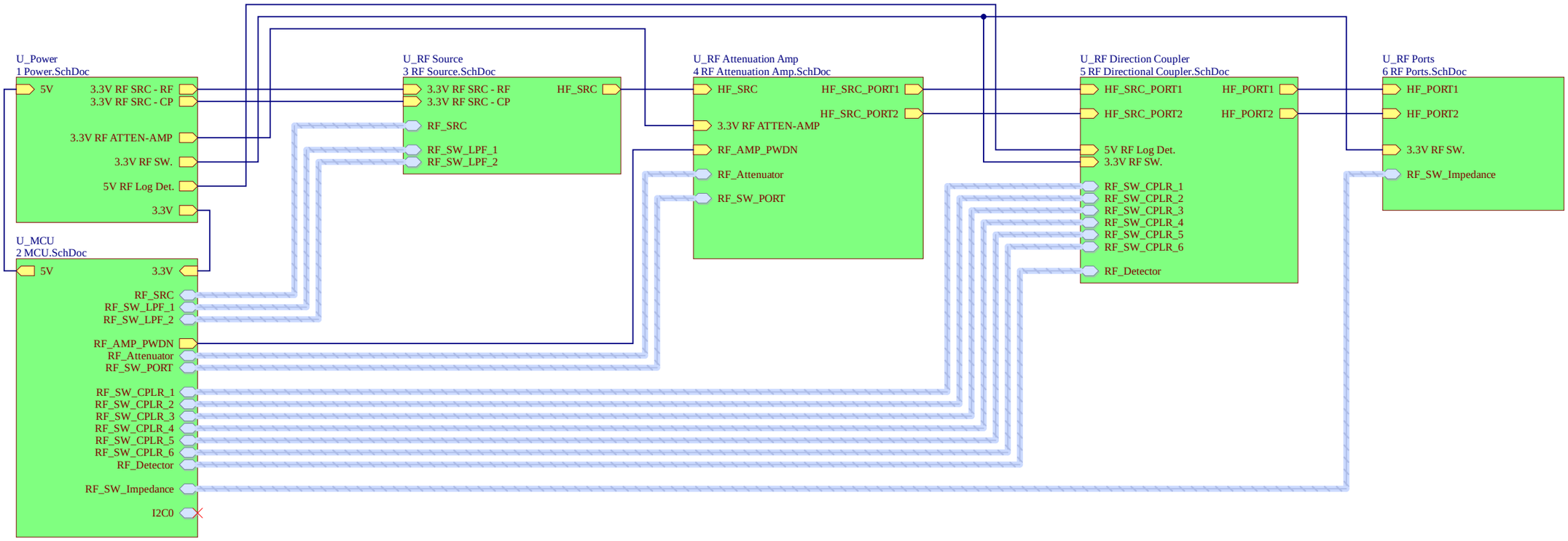
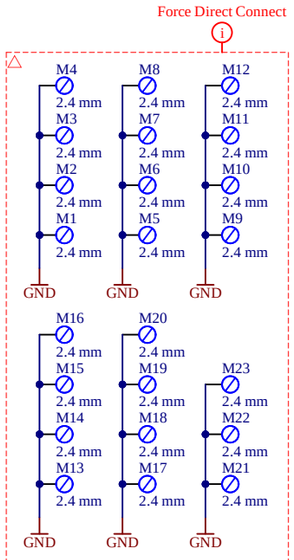
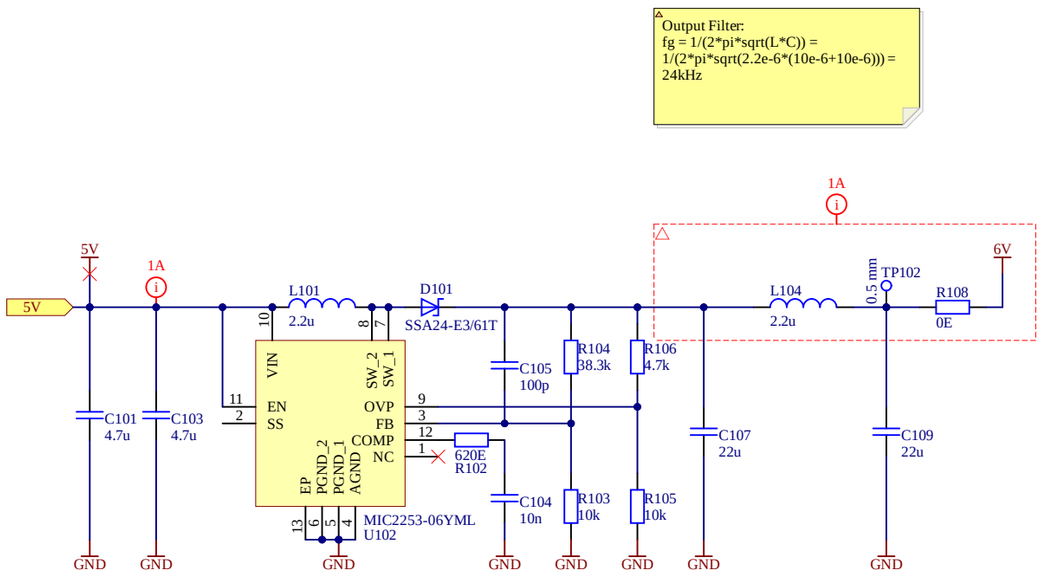


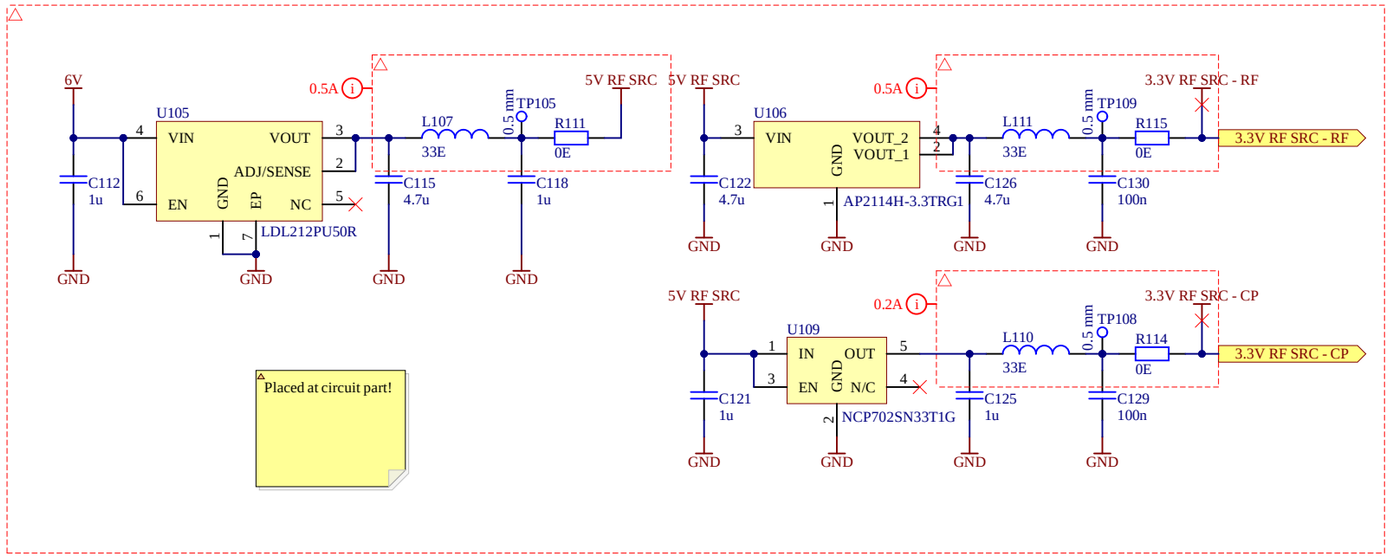


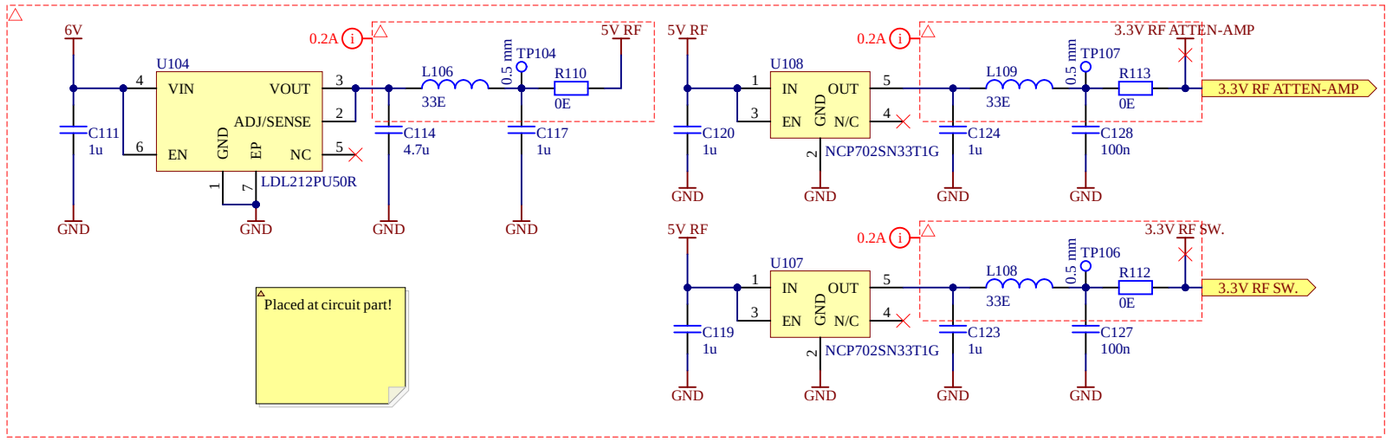
VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

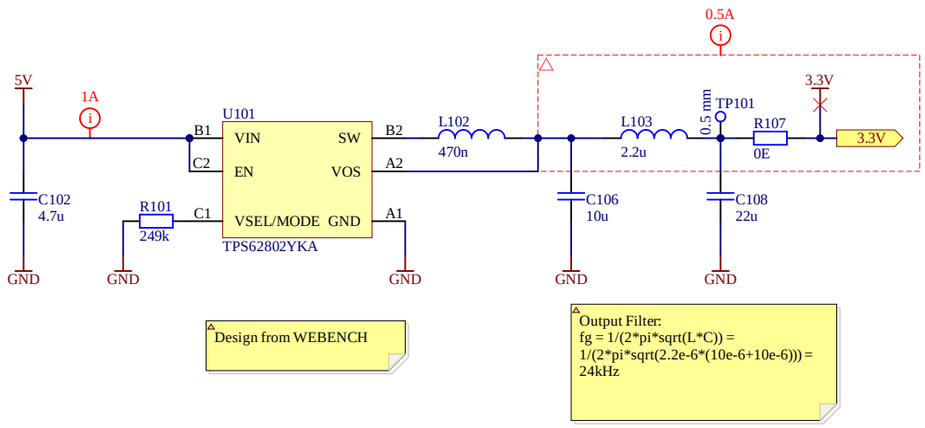
Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	**Project:** SNA **Board:** SNA **Page:** Overview	**Number:** SNA **Variant:** EMS **Release:** dd.mm.yyyy	**Sheet** 1 of 7 **Size:** A3	**Git:** 15d510e4 [Locally M **Version:** 1.0 **Design:** ML **Check:** ML

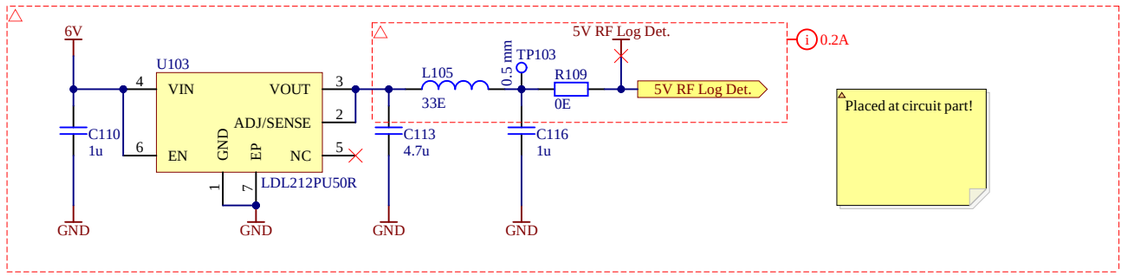
File: https://github.com/ml3rc/SNA.git 0 Overview.SchDoc







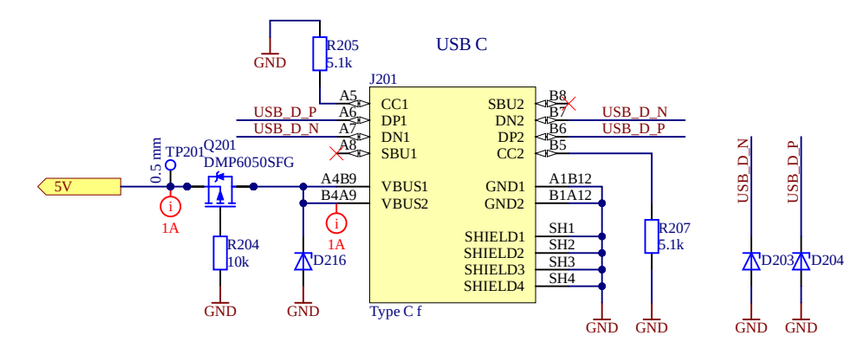


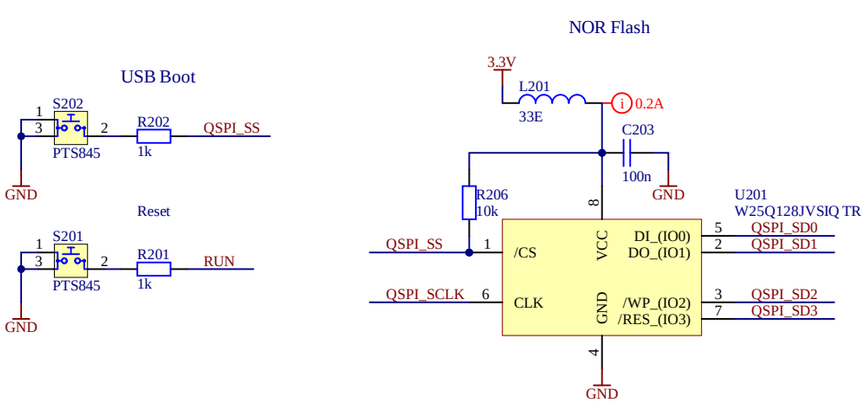


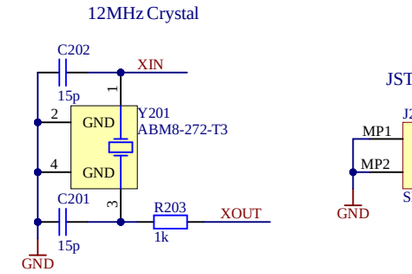
For current specs see power blockdiagramm
20mil -> 1A (2.14A)
15mil -> 0.5A (1.79A)
12mil -> 0.2A (1.56A)

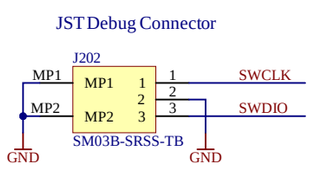
VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

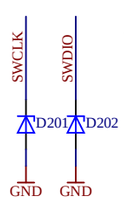
Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	**Project:** SNA **Board:** SNA **Page:** Power	**Number:** SNA **Variant:** EMS **Release:** dd.mm.yyyy	**Sheet** 2 of 7 **Size:** A3	**Git:** 0553af0a **Version:** 1.0 **Design:** ML **Check:** ML
		File: https://github.com/ml3rc/SNA.git		1 Power.SchDoc

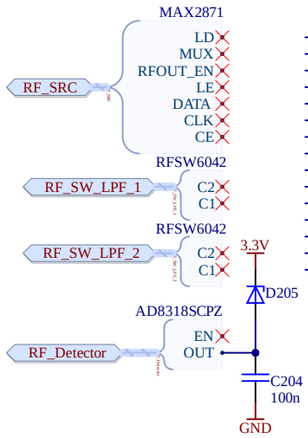


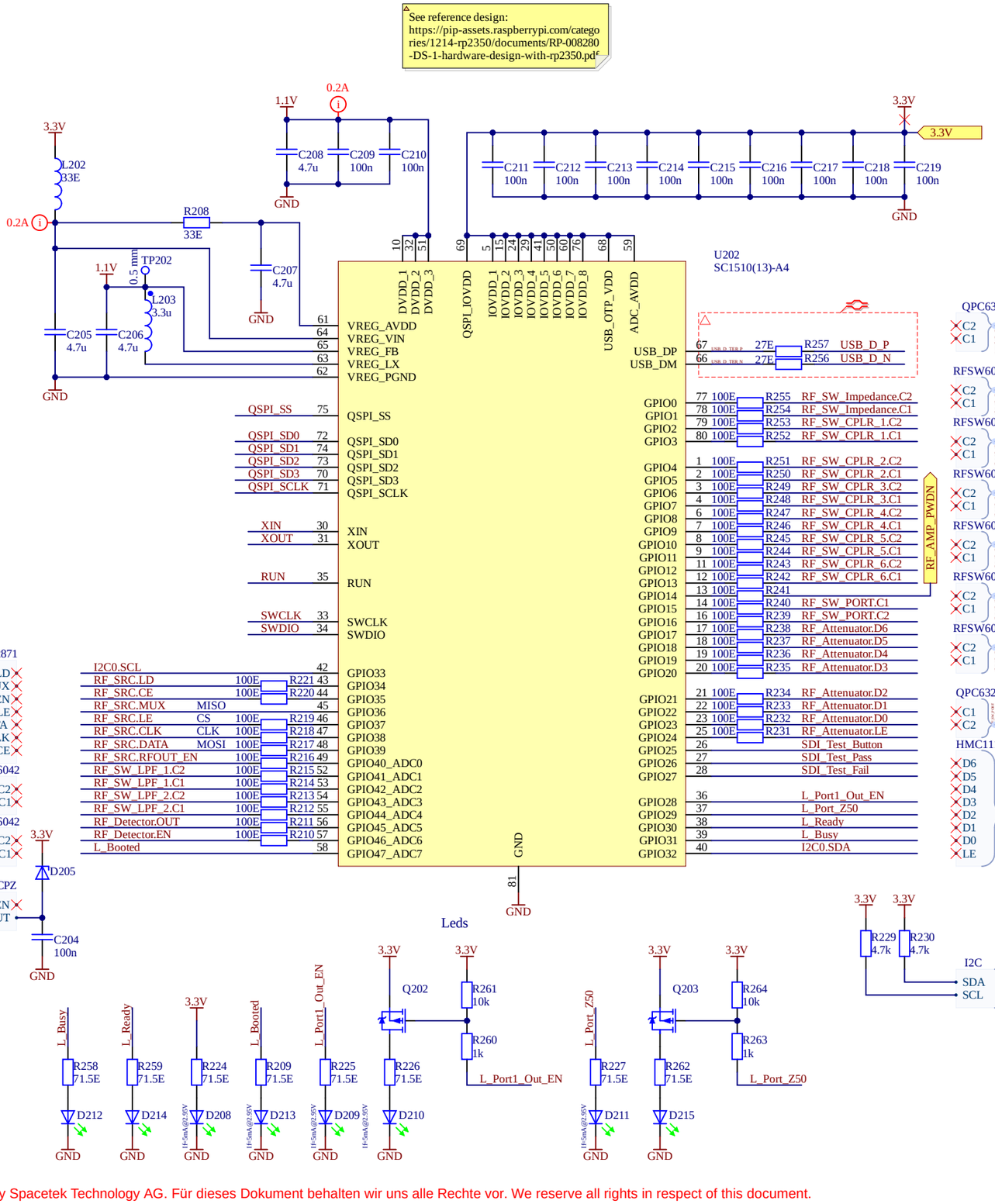


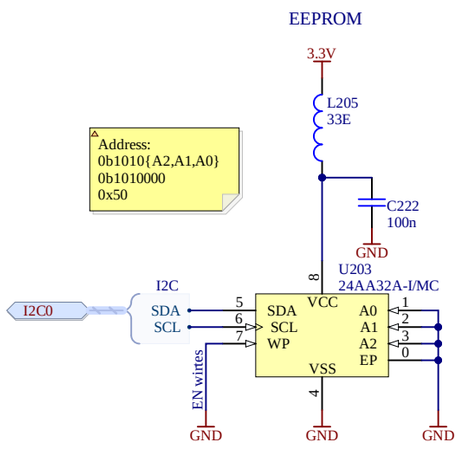


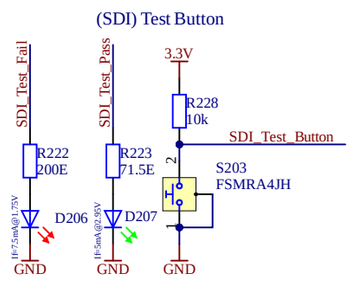


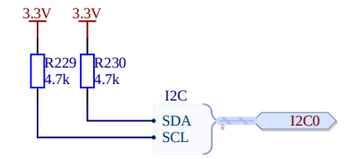










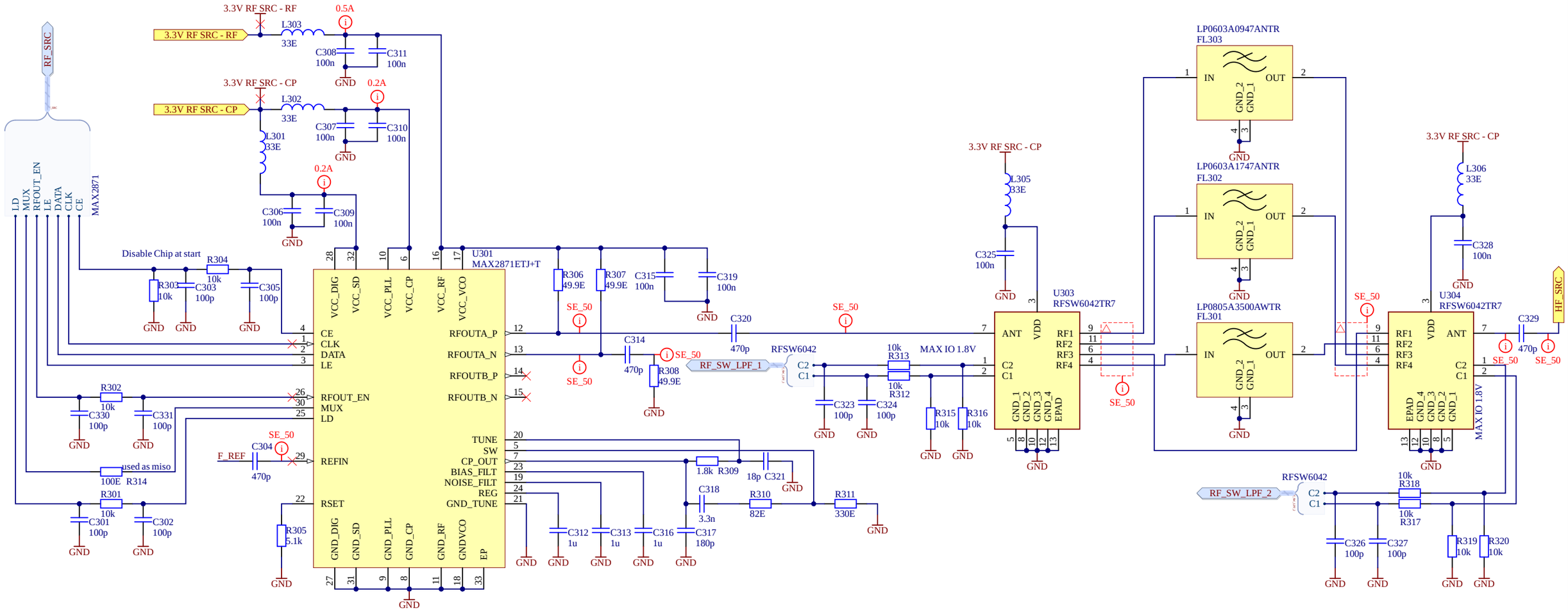


VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	**Project:** SNA **Board:** SNA **Page:** MCU	Number: SNA Variant: EMS Release: dd.mm.yyyy	Sheet 3 of 7 Size: A3	Git: fed1a5fe Version: 1.0 Design: ML Check: ML

File: https://github.com/ml3rc/SNA.git — 2 MCU.SchDoc

See LibreVNA Schematics for reference



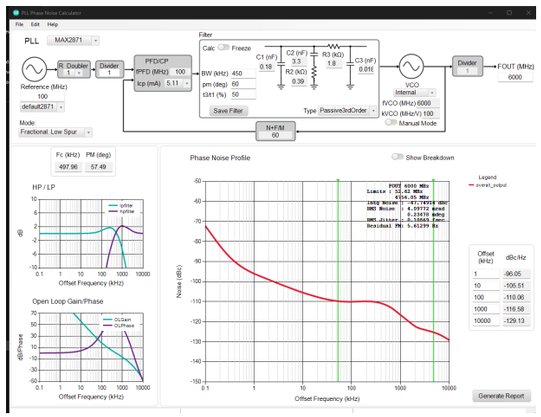
Rset = 5.1kOhm as per datasheet

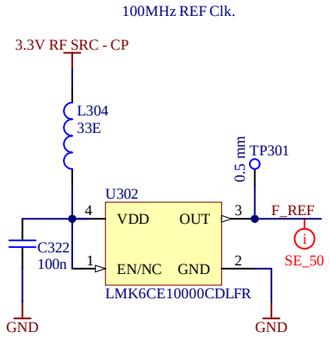
C_BIAS_FILT, C_NOISE_FILT and C_REG = 1uF as per datasheet

The loop filter can be calculated with:
https://www.analog.com/en/resources/evaluation-hardware-and-software/embedded-development-software/software-download.html?swpart=SFW0007420A

f_bandwidth_loopfilter = ~450kHz -> fast response, but higher noise

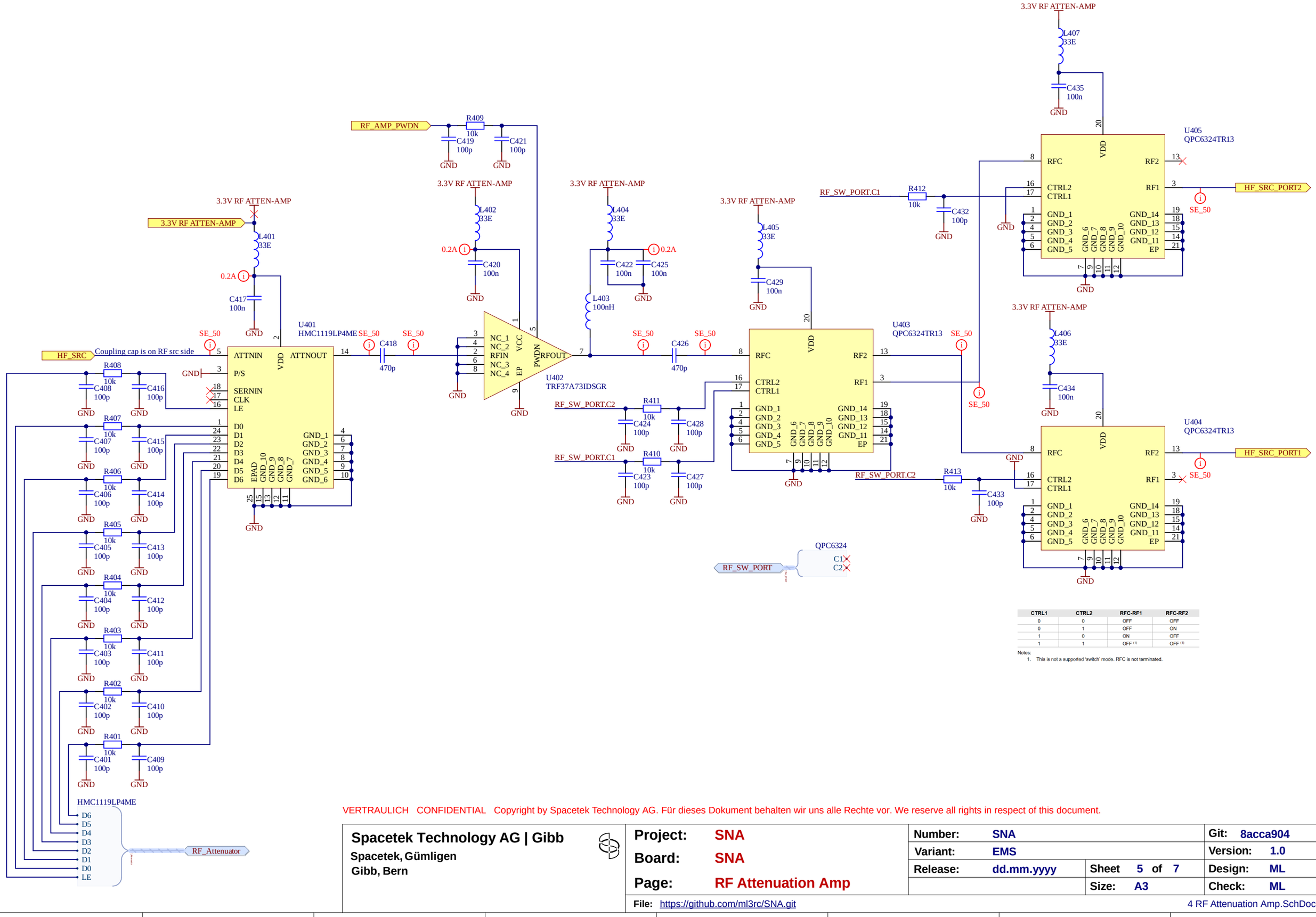
R_Crf_coupling = ((1)/(2*π*50MHz*470pF)) = 6.8Ohm





VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	**Project:** SNA **Board:** SNA **Page:** RF Source	**Number:** SNA **Variant:** EMS **Release:** dd.mm.yyyy	**Sheet** 4 of 7 **Size:** A3	**Git:** fd419390 **Version:** 1.0 **Design:** ML **Check:** ML
		File: https://github.com/ml3rc/SNA.git		3 RF Source.SchDoc



CTRL1	CTRL2	RFC-RF1	RFC-RF2
0	0	OFF	OFF
0	1	OFF	ON
1	0	ON	OFF
1	1	OFF (1)	OFF (1)

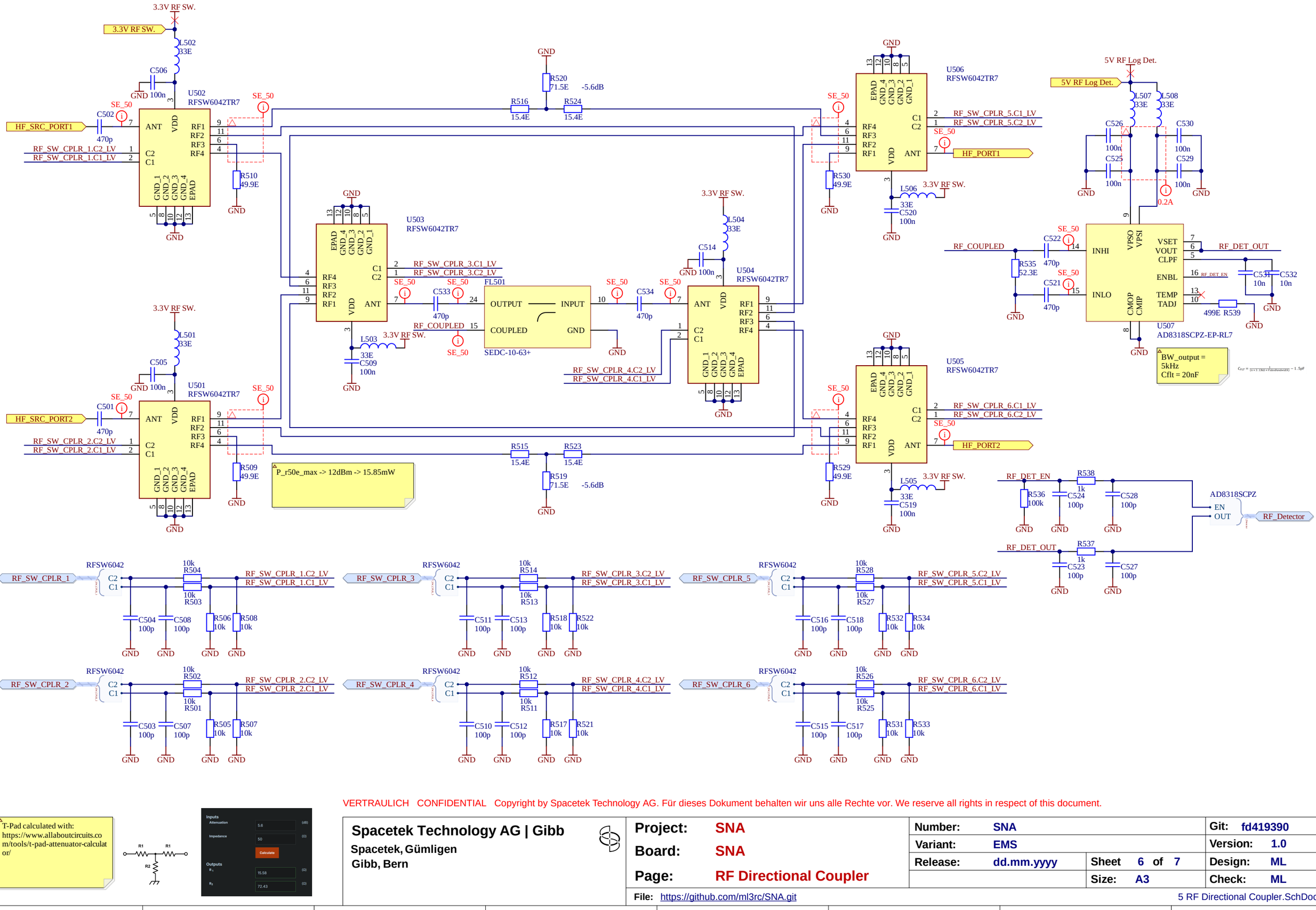
Notes:
1. This is not a supported 'switch' mode. RFC is not terminated.

VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

Spacetek Technology AG | Gibb
Spacetek, Gümligen
Gibb, Bern

Project:	SNA	**Number:**	SNA	**Git:**	8acca904
Board:	SNA	**Variant:**	EMS	**Version:**	1.0
Page:	RF Attenuation Amp	**Release:**	dd.mm.yyyy	**Sheet**	5 of 7
				Size:	A3
				Design:	ML
				Check:	ML

File: https://github.com/ml3rc/SNA.git 4 RF Attenuation Amp.SchDoc



T-Pad calculated with: https://www.allaboutcircuits.com/tools/t-pad-attenuator-calculator/

VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

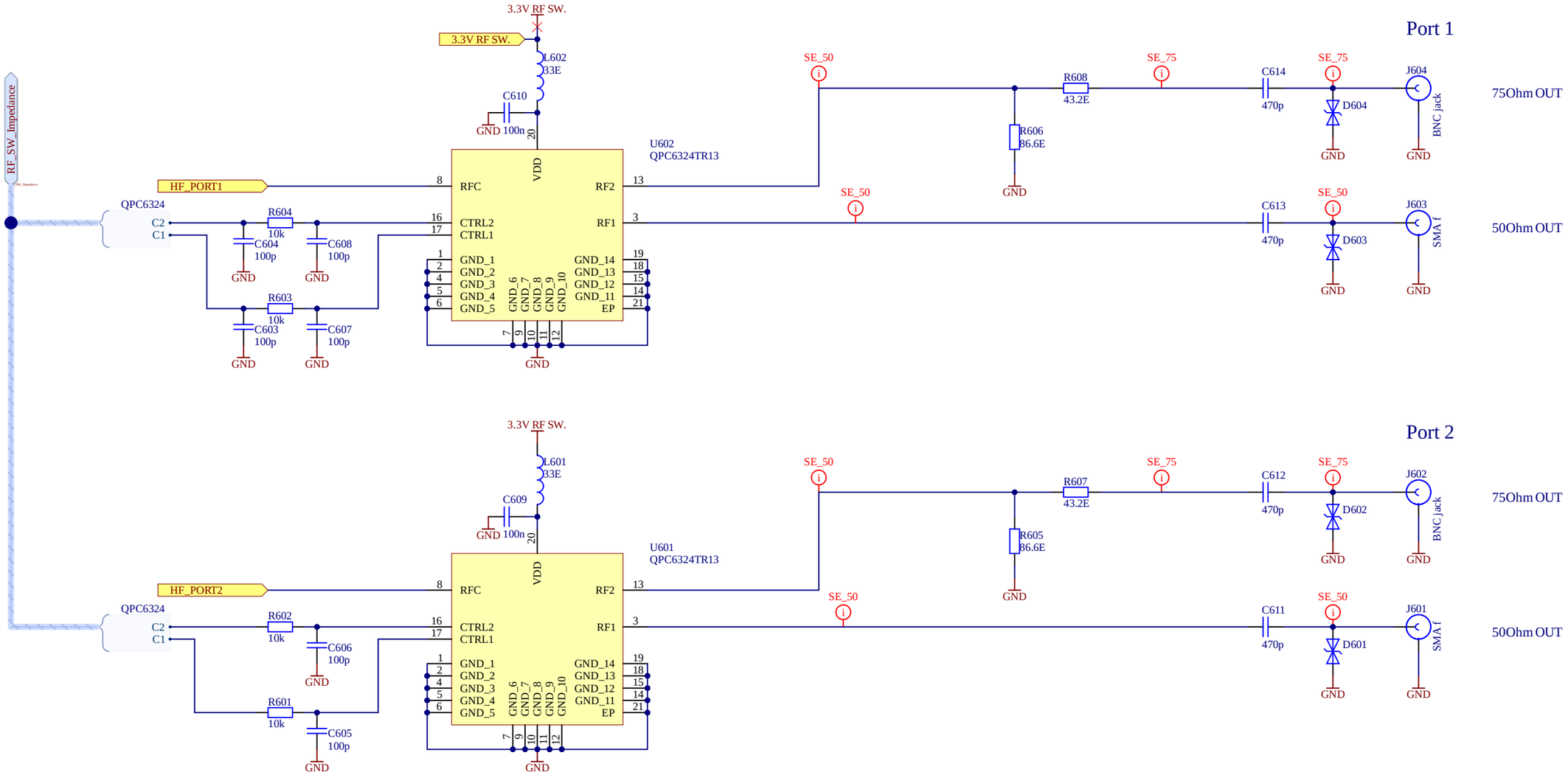
Spacetek Technology AG | Gibb

Spacetek, Gümligen

Gibb, Bern

Project:	**SNA**	**Number:**	**SNA**	**Git:**	**fd419390**
Board:	**SNA**	**Variant:**	**EMS**	**Version:**	**1.0**
Page:	**RF Directional Coupler**	**Release:**	**dd.mm.yyyy**	**Sheet**	**6 of 7**
				Size:	**A3**
				Design:	**ML**
				Check:	**ML**

File: https://github.com/ml3rc/SNA.git 5 RF Directional Coupler.SchDoc



50-75 Matching Pad:
solve({75=((1)/(((1)/(Rp))+((1)/(50))))+Rs,50=((1)/(((1)/(75+Rs))+((1)/(Rp))))})
Rp = 86.6Ohm
Rs = 43.3Ohm
Insertion Loss = 5.72dB

VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

Spacetek Technology AG | Gibb
Spacetek, Gümligen
Gibb, Bern

Project:	SNA	**Number:**	SNA	**Git:**	fed1a5fe
Board:	SNA	**Variant:**	EMS	**Version:**	1.0
Page:	RF Ports	**Release:**	dd.mm.yyyy	**Sheet**	7 of 7
				Design:	ML
				Size:	A3
				Check:	ML

File: https://github.com/ml3rc/SNA.git 6 RF Ports.SchDoc